

Safe System of Work

Operating the CDE 200tph C&D Waste Wash Plant

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Record of Amendments			

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Overview

This Safe System of Work sets out the company's Standard Operating Procedures for operating the CDE 200tph C&D waste wash plant at the J.A. Jackson Fullwood Recycling Facility, Preston. The plant is a wet processing system supplied by CDE Group that accepts construction & demolition (C&D) waste from the Fox Group tipper wagon fleet and produces washed sand and aggregate products (0–2 mm sand, 0–4 mm sand / grit, 10 mm, 10–20 mm, 20–40 mm aggregate and oversized aggregate) together with a filter cake by-product. Approximately 10,000 tonnes of washed material is produced per week and up to 85% of the feed is recycled into the local construction supply chain.

All employees, contractors and visitors have a responsibility for their safety and the safety of others by adhering to this SSoW alongside the, Start-Up Checklist (WP-01), Maintenance Checklist (WP-02) and the site Traffic Management Plan.

Plant Description

The plant is arranged as follows:

- Feed hopper with apron feeder receiving C&D material from wheel loader.
- Primary scalping screen removing oversize and directing material to the wash line.
- Picking station — manned belt to remove timber, plastic, rebar, tyres and hazardous items.
- Rinsing / sizing screen producing 10 mm, 10–20 mm and 20–40 mm washed aggregate and clean oversize for recycled Type 1 MOT production.
- Two EvoWash sand plants — one producing 0–4 mm sand / grit, the other 0–2 mm sand — each with a cyclone and dewatering screen.
- Water treatment via thickener with flocculant dosing; underflow processed through filter press producing filter cake.
- Filtrate return to a process-water buffer tank; make-up water added as required. Closed-loop water recycling reduces mains draw.
- Stockpile conveyors delivering finished product to stockpile ground for loader off-take.
- Motor Control Centre (MCC) with SCADA / HMI operator interface and full lock-out isolation stations.

Hazards Identified

Operating the wash plant could result in serious injury or fatality if the controls in this SSoW and the associated Risk Assessments are not followed. Key hazards include:

- Contact with moving parts — screens, cyclones, agitators, conveyors, coupling shafts.
- Entanglement in belt drums, drives and take-up assemblies.
- Fall from height — cyclone tower, screen decks, filter-press platforms.
- Slips, trips and falls from spillage, hoses and wet walkways.
- Collision with mobile plant — wheel loader, telehandler, tipper wagons.
- Drowning / engulfment in sumps, thickener, buffer tanks and cyclone underflow.
- Electrical shock at MCC and motor terminal boxes.
- Stored hydraulic and pneumatic energy in filter press, feed hopper and air receivers.
- Exposure to respirable crystalline silica (RCS) and other dusts.
- Noise exposure ≥ 85 dB in operating zones.
- Hand-arm and whole-body vibration.
- Manual handling of screen media, cyclone parts and filter-press cloths.
- Chemical exposure — flocculant, lubricants, coolants and hydraulic fluids.
- Fire from hydraulic oil, diesel, electrical fault or hot works.
- Adverse weather (high wind, lightning, freeze / thaw, heat stress).
- Environmental release — silt-laden run-off, oil spillage.
- Presence of unexpected items in the feed — asbestos, gas cylinders, chemical drums.
- Lone working during out-of-hours starts, wash-downs and fault-finding.

Personal Protective Equipment (PPE) Requirements

- Hi-vis jacket and trousers to EN ISO 20471.
- Safety helmet with chin-strap when at height or where struck-by risk exists.
- Safety footwear — steel toecap and midsole, slip-resistant sole to SRC.
- Impact-rated safety glasses (all operating areas).
- Ear defenders — SNR ≥ 27 dB — in mandatory hearing-protection zones.
- Cut-resistant gloves for handling screen media, cloths and sharps.
- FFP3 dust mask when dry-handling filter cake or working downwind of dust sources.
- Full-body harness (rated) with dual lanyards for temporary work at height outside fixed guardrails.
- Chemical gloves and splash goggles when handling flocculant, lubricants and coolants.

Pre-Use Checks

Before start-up each shift, the plant operator completes the WP-01 Start-Up Checklist. No item may be signed off if a defect is present. Any failed item is reported to the Plant Supervisor and the plant remains isolated until the defect is corrected.

- Confirm no permits, LOTO tags or isolations remain from the previous shift.
- Read the plant handover log and previous-shift maintenance log.
- Walk the plant perimeter and internal footprint; no unauthorised persons.
- Confirm water systems, hydraulic and grease reservoirs, and process fluids at correct level.
- Visually inspect all guards, e-stops and pull-cord trip wires.
- Test e-stops and rip-cords under the start-up sequence.

Operating Procedure

The following start-up sequence is strictly followed:

- Sound plant klaxon (3 long blasts) and wait 30 seconds before energising.
- Start process water — pumps and ring-main; confirm pressure stable.
- Start thickener rake and flocculant dosing.
- Set filter press to standby ready to receive underflow.
- Start stockpile conveyors from the discharge end back toward the plant.
- Start sand-plant dewatering screens and cyclone feed pumps.
- Start sizing and scalping screens.
- Start picking-station belt.
- Start apron feeder at minimum speed.
- Signal the wheel loader to begin feeding; ramp gradually to 200 tph target.

During operation:

- Operator monitors HMI for alarms; investigate any alarm before continuing to feed.
- Picking-station staff pick and clear the belt continuously; discard skip emptied before it overfills.
- Product stockpiles kept within permitted angle of repose; loader driver briefed on stockpile safe-work practice.
- Filter press cycled to timer and pressure setpoints; filter cake removed and stockpiled or reused.
- Any unexpected item at the picking station — asbestos-suspect, gas cylinder, chemical drum — trigger STOP: e-stop the belt, alert supervisor, isolate the plant, do not attempt to remove without a written safe method.

At shift end:

- Stop feed and allow material through the plant until screens and cyclones run empty.
- Stop feeders and picking belt.
- Stop screens, sand plants and stockpile conveyors in reverse order to start-up.
- Continue thickener and filter press until process water is clarified.
- Wash-down of picking station, screens and cyclone splash zones per the WP-02 daily maintenance list.
- Complete plant log — hours, tonnage, alarms, defects. Sign the handover log.

Isolation, Lock-Out and Permit Procedures

Any intervention beyond routine operation — clearing a blockage inside a guard, replacing screen media, changing filter cloths, inspecting cyclone internals, working inside a sump or on the thickener — is performed only after full LOTO in accordance with the Fox Group FoxSafe Isolation & Lock Out Procedure and under a signed Permit to Work.

- Every isolation point locked with the individual worker's LOTO padlock and personal danger tag.
- Zero-energy verified by attempted start, gauge reading and voltage test as applicable.
- Confined space entry (thickener, sumps, buffer tanks, hoppers) requires a separate Confined Space Permit, gas monitor and top-man.
- Hot works (welding, cutting, grinding) require a Hot Works Permit and fire watch during and after works.
- Working at height beyond fixed guardrails requires a rated harness anchored to an approved point, rescue plan and second-person.
- Never remove another worker's lock. If the worker is unavailable the Lock Removal Procedure signed off by the SHEQ Manager is followed.

Emergency Response

- Any injury or near-miss reported immediately to the Plant Supervisor and recorded in the incident log.
- First-aid provision — a competent first aider at the Fullwood site (see FoxSafe First Aid register).
- Fire — activate nearest fire point, evacuate to designated assembly point, dial 999 as required.
- Spill — deploy spill kit, contain the release, notify the SHEQ Manager and follow the Environmental Emergency Procedure.
- Entrapment — do not free-lift a casualty; e-stop, isolate energy, call for support and follow the rescue plan.

Training and Competence

- Only trained, competent and authorised operators may run the wash plant.
- Site induction and plant-specific SOP training completed before any solo operation.
- Refresher briefing after any significant plant modification, incident or SSoW revision.
- Toolbox talks on high-risk topics (silica, noise, LOTO, confined space) delivered at least six-monthly.
- Health surveillance — audiometry, HAVS, respiratory / RCS monitoring — for staff on the exposure register.

Related Documents

- WP-01 — Wash Plant Start-Up Checklist.
- WP-02 — Wash Plant Planned Maintenance Checklist.
- FoxSafe Isolation and Lock Out Procedure.
- Site Traffic Management Plan.
- Fox Group Permit to Work suite (General, Hot Works, Confined Space, Electrical).
- COSHH Register (flocculant, lubricants, coolants, hydraulic oils).
- Quarries Regulations 1999 (not applicable to C&D recycling but referenced for read-across).
- CDE OEM operating and maintenance manuals for feed hopper, screens, EvoWash sand plants, thickener and filter press.